Skittles diffusion

How to work as a group

Each group need only turn in one report. You should all discuss all of the questions and try to come up with answers as a group.

Quick overview of the experiment

The Skittles-diffusion experiment is a well-known experiment – feel free to Google "Skittles diffusion" and see what you get! Since the two main forces that drive bioelectricity are drift and diffusion, the experiment is also a good way to understand diffusion and start our journey into bioelectricity.

There are numerous Skittles-diffusion videos on YouTube; a nice short and snappy one (that even has nice music) is at https://www.youtube.com/watch?v=oDgNBkxDoPM .

Instructions

Grab yourself some fun-colored Skittles and some kind of a dish where you can pour water about half an inch deep. Drop in a few different-colored Skittles evenly spaced around the perimeter of the dish as in the video above, or pick your own favorite pattern.

After it seems to have reached a steady state, snap a picture of your beautiful result. Then answer the discussion questions and turn in the picture and your answers.

Discussion questions:

1. Include picture(s) showing your final result (40 points)
2. The final result, with color zones that have sharp boundaries, seems to defy diffusion. So: in what way does the lab actually demonstrate diffusion? (20 points)
3. Again, the final result, with color zones that have sharp boundaries, seems to defy diffusion. If you had not known this end result, what might you expect the eventual result of the experiment to be? (20 points)
4. You probably noticed that the colors initially spread quickly, and then the process slowed down. Though we didn't discuss this in class, it turns out that this behavior is well known – diffusing over twice the distance takes 4x as much time (in general, diffusing over a distance of $n*d$ takes n^2 times longer than diffusing over a distance d). Any idea why this might be, based on what we actually did discuss in class? (You can assume that pretty much all the dye dissolves quickly, and after that the dye just diffuses around). (15 points)
5. Can you speculate why the colors don't all mix together? This is actually a difficult question, and you'll find plenty of misinformation on the web. But take your best shot! (And don't worry if your best shot is wrong). (5 points)